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Abstract

We summarize the theoretical predictions of gravitational waves (GWs) in stellar core-collapse and core-collapse supernova evolution. Following a brief introduction to overview how scientifically significant the successful detection of GWs could be, we give a concise summary of the essential GW features mostly based on a back-of-the-envelope estimation. If the gravitational collapse of stellar cores and the subsequent explosion hydrodynamics occur in a perfectly spherically symmetric manner, no GWs can be emitted. Therefore, what makes the dynamics of the central engine deviate from spherical symmetry is essential for determining the physics of the GW emission processes. Among the candidates, we mainly focus on the best-studied GW signal that is emitted near core bounce in rapidly rotating core-collapse. After bounce, multiple emission processes have been proposed thus far. These postbounce GWs, if observed, are also expected to provide smoking-gun information for unraveling the yet uncertain explosion mechanisms. We finally conclude with the most urgent tasks to prepare for the GW astronomy of core-collapse supernovae.

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1 Introduction

On September 14, 2015, the centennial year after Albert Einstein’s 1915 general theory of relativity, the twin detectors of the Laser Interferometer Gravitational-Wave Observatory (LIGO) coined the first direct detection of gravitational waves (GWs). After the dedicated data analysis was done in order to unambiguously distinguish the real GW signal from detector noise, it was revealed that this Nobel Prize-winning event, GW150914, was produced by the merger of two black holes (BHs) (Abbott et al. 2016b). In addition to LIGO, the second-generation detectors such as Advanced VIRGO (Hild et al. 2009) and KAGRA (Aso et al. 2013) will be on line in the coming years. Furthermore, a third-generation detector like the Einstein telescope has also recently been proposed (Punturo et al. 2014). At such a high level of precision, these detectors are sensitive enough to detect a wide variety of compact objects.

The prime targets are compact binary coalescences such as the merger of black holes and/or neutron stars (NSs) (e.g., Sathyaprakash and Schutz 2009). Other intriguing sources include neutron star normal mode oscillations and rotating neutron star mountains (e.g., Andersson 2003). Core-collapse supernovae (CCSNe) are the mothers of all these compact objects. For massive stars heavier than $\sim 8M_{\odot}$ (Smartt 2015), the central iron core, once formed at the final stage of stellar evolution, is dynamically unstable and begins gravitational collapse (Iben 2013). The focus of this chapter is on the GW emission produced in the blink of an eye (in approximately one second, contrasting with the million years of stellar evolution) when the catastrophic implosion turns into one of the most energetic explosions in the modern Universe.

According to Einstein’s theory of general relativity (GR, e.g., Misner et al. 1973), no GWs can be emitted if gravitational collapse of the stellar core proceeds with perfect spherical symmetry. To produce GWs, the central engine that powers the explosion should work aspherically and dynamically. In fact, there is accumulating observational evidence from electromagnetic-wave observations of ejecta morphologies, spatial distributions of nucleosynthetic yields, and natal kick of pulsars, all of which have pointed towards CCSNe being generally aspherical (i.e., spatially multidimensional (multi-D), e.g., Wang et al. (2002), Maeda et al. (2008), Tanaka et al. (2009), Grefenstette et al. (2014), and references therein). Unfortunately, however, these electromagnetic signatures are secondary for probing the inner workings of CCSNe because they are only able to provide images of optically thin regions far away from the central core.

Much more direct information is delivered to us by neutrinos and GWs. In fact, the detection of neutrinos from SN 1987A opened the door for neutrino astronomy. (See, e.g., ► Chap. 59, “Neutrino Emission from Supernovae”.) However, these neutrino signatures produced deep inside the core are influenced, in propagating through the stellar envelope, by both collective neutrino oscillations and the Mikheyev-Smirnov-Wolfenstein effects. (See, e.g., ► Chap. 59, “Neutrino Emission from Supernovae”.) Therefore, GWs are the most direct observable bringing us news of this intimate episode of the supernova engine. Current estimates of CCSN rates in our Milky Way predict one event every $\sim 40 \pm 10$ years (Horiuchi et al. 2013). Although rare, they provide a unique way to study CCSNe using a full set of multimessenger observables including GWs, neutrinos, and electromagnetic waves (e.g., Nakamura et al. 2016) including nuclear gamma-rays (Gehrels et al. 1987).

From a theoretical point of view, understanding the origin of the explosion multidimensionalities is also indispensable for clarifying the yet uncertain CCSN mechanisms. After a half-century of extensive research, theory and neutrino radiation hydrodynamics simulations are now reaching a point where multi-D hydrodynamic instabilities play a crucial role in the neutrino-heating mechanism of CCSNe (Bethe 1990), the most favored scenario to trigger explosions. In fact, a number of first-principle, self-consistent simulations in two or three spatial dimensions (2D, 3D) now report successful neutrino-driven models that are trending towards explosion. These successful models have strengthened our confidence in the multi-D neutrino-driven paradigm (see Janka et al. (2016), Foglizzo et al. (2015), Burrows (2013), and Kotake et al. (2012b) for recent reviews).

Another candidate mechanism is the magnetohydrodynamic (MHD) mechanism (e.g., LeBlanc and Wilson 1970, Kotake et al. 2006, Thielemann et al. 2012, Mösta et al. 2014, and Nishimura et al. 2015). Rapid rotation of precollapse iron cores is necessary for this mechanism, because it relies on the extraction of rotational free energy of the core by means of field-wrapping and magnetorotational instability (MRI, e.g., Balbus and Hawley 1998, Obergaulinger et al. 2009, Masada et al. 2012, and references therein). Such rapid rotation is likely to be typical of $\sim 1\%$ of the massive star population (e.g., Woosley and Bloom 2006). Minor as they may be, the MHD explosions are attracting great attention as they are possibly relevant for magnetars and collapsars (e.g., Woosley 1993 and Metzger et al. 2011), which are hypothetically linked to the formation of long-duration gamma-ray bursts (e.g., Mészáros (2006) for a review).

In step with these advances in the CCSN theory, significant progress in understanding GW emission processes has been made almost at the same pace (e.g., Kotake (2013), Fryer and New (2011), and Ott (2009) for recent reviews). In the MHD mechanism, rapid rotation of the precollapse core leads to significant rotational flattening of the collapsing and bouncing core, leading to a waveform that has been theoretically best-studied, the so-called type I waveform of the bounce GW signal. The waveform is characterized by sharp spikes at bounce followed by a subsequent ring-down phase (Dimmelmeier et al. 2007; Kotake et al. 2003; Mönchmeyer et al. 1991; Ott et al. 2004, 2007). After bounce, a large variety of emission processes has been proposed, including convective motions in the

proto-neutron star (PNS) and in the region behind the stalled shock (Burrows and Hayes 1996; Müller and Janka 1997; Müller et al. 2004), the standing-accretion-shock-instability (SASI; e.g., Kotake et al. 2009, 2007; Marek et al. 2009; Murphy et al. 2009), nonaxisymmetric rotational instabilities (Kuroda et al. 2014; Ott et al. 2005; Scheidegger et al. 2010), and anisotropy in neutrino emission (Kotake et al. 2009; Müller et al. 2013, 2012, 2004).

This chapter is structured as follows. In Sect. 2, we give a brief summary of the essential GW features in CCSNe mostly based on a back-of-the-envelope estimation. We primarily focus on the well-established GW signal emitted near core bounce in rapidly rotating core-collapse (Sect. 3). Regarding the GWs in the postbounce phase, we attempt in Sect. 4 to select only the established GW signatures from the various emission mechanisms proposed thus far. In Sect. 5, we give a summary and discuss some urgent tasks to prepare for the upcoming era of the GW astronomy of CCSNe.

2 Basic GW Features

In this section, we give a brief summary of the fundamental GW signatures of CCSNe. For physical foundations of GWs, the reader is referred to standard textbooks (e.g., Misner et al. 1973; Shapiro and Teukolsky 1983) for a complete description. Instead of re-expressing them, here we only choose several formulae useful in the CCSN context and present order-of-magnitude estimates of the GW amplitudes and frequencies for later convenience.

2.1 GW Extraction Formulae

2.1.1 Quadrupole Formula

As we later show, GWs emitted in stellar core-collapse have little impact on the evolution of the hydrodynamics. Due to the weakness of the signal, the quadrupole formula (e.g., Misner et al. 1973), which is the simplest approach to extract the gravitational waveform, has been widely used in the community. The validity of the approximation has been confirmed by comparing with more detailed GW extraction techniques based on numerical relativity (Reisswig et al. 2011).

In the quadrupole formalism (e.g., Section 16 in Shapiro and Teukolsky 1983), the dimensionless GW strain h is

$$h_{ij}^{TT}(t, \mathbf{x}) = \frac{2G}{c^4 D} \ddot{I}_{ij}^{TT} \left(t - \frac{D}{c} \right), \quad \left(\equiv \frac{\partial}{\partial t} \right), \quad (1)$$

where $D = |\mathbf{x}|$ is the distance to the source, G is the gravitational constant, and c is the speed of light. I_{ij}^{TT} is the transverse-traceless (TT) part of the mass quadrupole moment I_{ij} ,

$$I_{ij} = \int d^3x \rho(x_i x_j - \frac{1}{3} \delta_{ij} \mathbf{x}^2), \quad (2)$$

where ρ is the mass density. The energy flux from the quadrupole GW emission is given by the Isaacson stress-energy tensor as $T_{0r} \propto \langle h_{ij,0}^{TT} h_{ij,r}^{TT} \rangle$, where r represents the radial direction to the source and the angle brackets denote appropriately averaging processes (such as over several wavelengths). Then one can obtain the energy loss rate, equivalently the GW luminosity, by integrating the energy flux over a distant sphere (see the above textbooks for the derivation),

$$L_{\text{GW}} = \frac{G}{5c^5} \langle \ddot{I}^{ij} \ddot{I}_{ij} \rangle_{TT}. \quad (3)$$

Using Eqs. (1) and (3), we give an order-of-magnitude estimate of the GW amplitude and luminosity emitted near core bounce. The GW amplitude can be estimated as

$$|h| = \frac{2G}{c^4 D} \ddot{I}_{ij} \sim \frac{2G}{c^4 D} \frac{MR_c^2}{T_{\text{dyn}}^2} \simeq \frac{300 \text{ cm}}{D} \epsilon \left(\frac{M}{M_\odot} \right) \left(\frac{R_c}{10 \text{ km}} \right)^2 \left(\frac{T_{\text{dyn}}}{1 \text{ ms}} \right)^{-2} \quad (4)$$

$$\simeq 10^{-21} \left(\frac{\epsilon}{0.1} \right) \left(\frac{10 \text{ kpc}}{D} \right) \left(\frac{M}{M_\odot} \right) \left(\frac{R_c}{10 \text{ km}} \right)^2 \left(\frac{T_{\text{dyn}}}{1 \text{ ms}} \right)^{-2}, \quad (5)$$

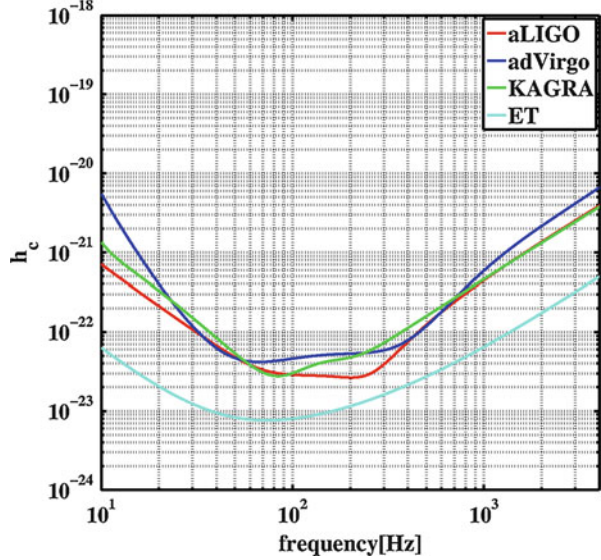
where $\ddot{I}_{ij}$ is now decomposed into the typical mass (M), radius of the (inner) core (R_c), and dynamical timescale at core bounce (T_{dyn}), respectively. We assume that the supernova occurs near our Galactic center at a distance of 10 kpc. ϵ is a parameter, representing the degree of the nonsphericity ($= 0$ if spherical), which may be optimistically estimated to be on the order of 10% in rapidly rotating supernova cores.

The typical GW frequency can be approximately estimated by the inverse of the dynamical timescale ($T_{\text{dyn}} \propto 1/\sqrt{G\rho} \sim 1 \text{ ms } \rho_{13}^{-1/2}$ with ρ being the average density in units of $10^{13} \text{ g cm}^{-3}$). Because the GW emission sites are mainly located in the vicinity of the proto-neutron star (PNS) ($10^{11} \text{ g cm}^{-3} \lesssim \rho \lesssim 10^{13} \text{ g cm}^{-3}$), the GW frequency is expected to be in the following range,

$$f_{\text{GW}} \sim \frac{1}{T_{\text{dyn}}} \simeq O(100 \text{ Hz}) \sim 1 \text{ kHz}. \quad (6)$$

Figure 1 summarizes the sensitivity curves of LIGO (labeled aLIGO in the figure; e.g., Abbott et al. 2016a), advanced Virgo (adVirgo; Hild et al. 2009), KAGRA (Aso et al. 2013), and the Einstein telescope (ET; Punturo et al. 2014). Plotted lines correspond to the dimensionless rms strain sensitivity $h_{\text{rms}} = \sqrt{fS(f)}$ where $S(f)$ is the detector-dependent noise power spectral density (for more details on the GW detectors, e.g., see ► Chap. 64, “Detecting Gravitational Waves from Supernovae with Advanced LIGO”). From Eqs. (5) and (6), one could see that a CCSN event in

Fig. 1 Sensitivity curves of representative laser interferometers. Currently (as of April 2016), LIGO (labeled aLIGO in the figure) is in operation, whereas advanced Virgo (adVirgo) and KAGRA are under construction, and the construction of the Einstein telescope is being proposed. See text for definition of the ordinate axis



our Galactic center ($D = 10$ kpc) is the primary target where the star formation rate should be highest in the Milky Way. When the ET is on line, the detection horizon could extend to megaparsec distance scales ($D \sim \text{Mpc}$). Considering the CCSN rate in the local universe (Horiuchi et al. 2013), one could optimistically expect the GW detection from CCSNe to be several per decade in the ET era.

Regarding the GW luminosity, the third time derivate of I_{ij} in Eq. (3) is approximately expressed as

$$\ddot{I}_{ij} \sim \epsilon M R_c^2 T_{\text{dyn}}^{-3} \sim \epsilon \left(\frac{c^5}{G} \right) \left(\frac{GM}{R_c c^2} \right) \left(\frac{v}{c} \right)^3, \quad (7)$$

where $v \sim R_c/T_{\text{dyn}}$ is the typical velocity in the core. Assuming energy conservation ($GM/R_c \sim v^2$), Eq. (3) becomes

$$L_{\text{GW}} \lesssim \epsilon^2 L_G \left(\frac{v}{c} \right)^{10}, \quad (8)$$

where

$$L_G = \frac{c^5}{G} = 3.6 \times 10^{59} \text{ erg s}^{-1}. \quad (9)$$

By taking the typical velocity ($v/c \sim 10\%$) and the duration ($\Delta t_{\text{exp}} \sim 1$ s) relevant for the CCSN engine, the total energy carried away by GWs can be estimated as $E_{\text{GW}} \sim L_{\text{GW}} \times \Delta t_{\text{exp}} \lesssim 10^{47} \epsilon_{0.1}^2 \text{ erg}$. This is really a crude upper bound. But, it is instructive to see that E_{GW} is much smaller compared to the total energy emitted in

the form of neutrinos ($E_\nu \sim 10^{53}$ erg) and photons ($E_{\text{photon}} \sim 10^{49}$ erg). Comparing this with the typical observed supernova kinetic energy ($E_K \sim 10^{51}$ erg, e.g., Tanaka et al. 2009), one may readily see that the GW emission is dynamically insignificant as mentioned earlier.

2.2 Axisymmetric Versus Nonaxisymmetric Sources

We move on to perform another simple estimate for the GW emission from the rotating star in axisymmetry (i.e., 2D). Then we present several useful formulae to extract the waveform in both 2D and 3D CCSN models.

In order to discuss the detectability of GWs seen from the distant observer, it is convenient to find the nonzero components of h_{ij}^{TT} in spherical coordinates, not in Cartesian coordinates. We assume that the GW propagates along the r direction from the source at a distance of D . Due to the TT nature, h_{rr} , $h_{r\theta}$, and $h_{r\phi}$ vanish. Then the $+$ and $\times$ mode of the GWs can be defined from the two nonvanishing components as

$$h_+ = \frac{h_{\theta\theta}}{D^2}, \quad h_\times = \frac{h_{\theta\phi}}{D^2 \sin \theta}. \quad (10)$$

By performing a simple coordinate transformation, for example, for the $\theta\theta$ component, $h_{\theta\theta} = h_{ij}^{TT} \frac{\partial x^i}{\partial \theta} \frac{\partial x^j}{\partial \theta}$, the two components read

$$\begin{aligned} \frac{h_{\theta\theta}}{D^2} &= \left[(h_{xx}^Q - h_{yy}^Q) \frac{(\cos^2 \theta + 1)}{4} \cos 2\phi - \frac{h_{xx}^Q + h_{yy}^Q - 2h_{zz}^Q}{4} \sin^2 \theta + \right. \\ &\quad \left. h_{xy}^Q \frac{\cos^2 \theta + 1}{2} \sin 2\phi - h_{xz}^Q \sin \theta \cos \theta \cos \phi - h_{yz}^Q \sin \theta \cos \theta \sin \phi \right], \\ \frac{h_{\theta\phi}}{D^2 \sin \theta} &= \\ &\quad - \frac{h_{xx}^Q - h_{yy}^Q}{2} \cos \theta \sin \phi + h_{xy}^Q \cos \theta \cos 2\phi + h_{xz}^Q \sin \theta \sin \phi - h_{yz}^Q \sin \theta \cos \phi, \end{aligned} \quad (11)$$

where $h_{ij}^Q = \frac{2}{D} \ddot{I}_{ij}^{TT}$ is expressed in Cartesian coordinates.

Using cylindrical coordinates $(\hat{R}, \hat{Z})$, the density distribution of an axisymmetrically rotating star is described as $\rho(\hat{R}, \hat{Z})$. The quadrupole moments are $I_{xx} = \pi \int \rho(\hat{R}, \hat{Z}) \hat{R}^3 d\hat{R} d\hat{Z}$, $I_{yy} = I_{xx}$, $I_{xy} = 0$. Furthermore, assuming equatorial symmetry of the rotating star, the remaining components are $I_{zz} = 2\pi \int \rho(\hat{R}, \hat{Z}) \hat{Z}^2 \hat{R} d\hat{R} d\hat{Z}$, $I_{xz} = I_{yz} = 0$. As easily understood, if the star rotates stationary, no GWs are emitted because $\ddot{I}_{ij} = 0$. The GWs are emitted when the rotating star contracts or expands dynamically, because the time derivatives of the

quadrupole moments have nonzero values. Note also that the GWs can be emitted from “axisymmetrically” rotating stars, when the motion is dynamically changing.

Inserting Eqs. (11) and (12) into Eq. (10), one can get

$$h_+ = -\frac{1}{D}(\ddot{I}_{xx} - \ddot{I}_{zz}) \sin^2 \theta, \quad (13)$$

and

$$h_\times = 0, \quad (14)$$

respectively, for $\phi = 0$.

From an axisymmetric source, the GW is most strongly emitted in the direction perpendicular to the symmetry axis ($\theta = \pi/2$ in Eq. (13)) and no GW is emitted toward the symmetry axis. It is intuitively natural because the dynamical behavior of the rotating star, if axisymmetric, can be seen most dramatically from the equatorial plane. As a side remark, GWs from merging compact objects (NSs or BHs) are generally most strongly emitted in the direction of the rotational axis.

In practice, the numerical treatment of the second time derivatives in Eq. (13) is formidable. Moreover, the use of spherical coordinates is most common in stellar core-collapse simulations (because a nonrotating star in equilibrium takes a spherical shape). Using the hydrodynamic equations of perfect fluids, the time derivatives can be eliminated. For example, the first time derivative of I_{xx} in Eq. (13) contains $\frac{\partial \rho}{\partial t}$ in the integrand. Using the equation of mass conservation expressed in spherical coordinates,

$$\frac{\partial \rho}{\partial t} + \frac{1}{r^2} \frac{\partial}{\partial r}(r^2 \rho v_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta}(\sin \theta \rho v_\theta) = 0 \quad (15)$$

with $\mu = \cos \theta$ and $\mathbf{v}$ being the velocity, the time derivative in the integrand can be replaced by the quantities with spatial derivatives. Taking one more time derivative using the Euler equations, one can get the most commonly used GW (quadrupole) formula in 2D (axisymmetric) CCSN simulations as

$$h_+ = \frac{1}{8} \sqrt{\frac{15}{\pi}} \sin^2 \theta \frac{A_{20}^{E2}}{D}, \quad (16)$$

where A_{20}^{E2} (the quadrupole amplitude, see the definition in Thorne 1980) is given as

$$\begin{aligned} A_{20}^{E2} = & \frac{G}{c^4} \frac{32\pi^{3/2}}{\sqrt{15}} \int_0^1 d\mu \int_0^\infty r^2 dr \rho [v_r^2 (3\mu^2 - 1) + v_\theta^2 (2 - 3\mu^2) - v_\phi^2 \\ & - 6v_r v_\theta \mu \sqrt{1 - \mu^2} - r \partial_r \Phi (3\mu^2 - 1) + 3\partial_\theta \Phi \mu \sqrt{1 - \mu^2}], \end{aligned} \quad (17)$$

where $\partial_r = \partial/\partial r$, $\partial_\theta = \partial/\partial\theta$ and Φ is the gravitational potential (see Mönchmeyer et al. (1991) for the derivation).

The spectral energy distribution dE_{GW}/df of the GW is given by

$$\frac{dE_{\text{GW}}}{df} = \frac{c^3}{16\pi G} (2\pi f)^2 \int_{-\infty}^{\infty} e^{-2\pi i f t} A_{20}^{\text{E2}}(t) dt, \quad (18)$$

then the gravitational-wave energy E_{GW} emitted in the GW is

$$E_{\text{GW}} = \int_0^{\infty} \frac{dE_{\text{GW}}}{df} df \quad (19)$$

Without axisymmetry in the supernova core, the $\times$ mode of GW (Eq. (14)) is no longer zero. The angle-dependent GW formulae usable in 3D (nonaxisymmetric) simulations were derived by Oohara et al. (1997) and compactly summarized in Müller et al. (2012) and Kuroda et al. (2014).

To access the detectability of GWs by interferometers, the following quantities are often used. One can calculate the spectral GW density $h_{\text{rss}} [\text{Hz}^{-1/2}]$ from the Fourier transform of the wave signals ($h_{+, \times}(f) = \int_{-\infty}^{\infty} e^{2\pi i f t} h_{+, \times}(t) dt$) as $h_{\text{rss}} = \sqrt{\sum_{A=+, \times} |\tilde{h}_A(f)|^2}$. Then the (detector-dependent) characteristic frequency is

$$f_c = \left(\int_0^{\infty} \frac{\sum_A \tilde{h}_A(f) \tilde{h}_A^*(f)}{S_n(f)} f df \right) / \left(\int_0^{\infty} \frac{\sum_A \tilde{h}_A(f) \tilde{h}_A^*(f)}{S_n(f)} df \right), \quad (20)$$

where $S_n(f)$ is the detector noise power spectral density in units of $\text{Hz}^{-1/2}$. Having the same unit as the strain equivalent spectrum density, h_{rss} is often used in burst GW searches to compare the signal strength with the detector sensitivity. h_c is the characteristic strain amplitude (Thorne 1987),

$$h_c \equiv \left(\int_0^{\infty} \frac{S_n(f_c)}{S_n(f)} \sum_A \tilde{h}_A(f) \tilde{h}_A^*(f) f df \right)^{1/2}, \quad (21)$$

by which one can compute the signal-to-noise ratio as

$$\text{signal-to-noise ratio} = \frac{h_c}{\sqrt{f_c S_n(f_c)}}. \quad (22)$$

To claim detection, a signal-to-noise ratio greater than unity and probably in the range of 8–13 would be needed (Flanagan and Hughes 1998; Sathyaprakash and Schutz 2009).

2.2.1 GWs from Anisotropic Neutrino Radiation

In addition to the quadrupole matter motions (Sect. 2.1.1), any sources that can contribute to the nonspherical part of the energy momentum tensor of the Einstein equations lead to GW emission. One of these sources is GW from anisotropic neutrino radiation emitted from the PNS. Epstein (1978) was the first to derive the formula of GWs produced by anisotropically escaping neutrinos from the CCSN core. One should define a concrete form of an energy-momentum tensor of the neutrino radiation field to compute the gravitational waves. In Epstein (1978), the following form is naturally assumed,

$$T_{\nu ij}(t, \mathbf{x}) = n_i n_j D^{-2} L_\nu(t-r) f(\Omega, t-r), \quad (23)$$

where $\mathbf{n}_i = \mathbf{x}_i/D$, $D = |\mathbf{x}|$, $f(\Omega, t) \geq 0$, and $\int f(\Omega, t) d\Omega = 1$. This source tensor represents radiation fields of neutrinos being released at the speed of light from the point $\mathbf{x} = 0$ to an observer at a distance of D . The functions $L_\nu(t)$ and $f(\Omega, t)$ are the rate of energy loss and the angular distribution of neutrino radiation, respectively, at time t . Given the energy-momentum tensor, one can calculate the TT part of the GW strain from neutrinos (Burrows and Hayes 1996; Müller and Janka 1997) as

$$h_{\nu ij}^{TT}(t, \mathbf{x}) = \frac{4G}{c^4 D} \int_{-\infty}^{t-D/c} \int_{4\pi} \frac{(n_i n_j)^{TT} L_\nu(t') f(\Omega', t')}{1 - \cos \theta'} d\Omega' dt', \quad (24)$$

where dashed (') variables represent the quantities observed in a coordinate frame (x' , y' , z'); on the other hand, nondashed variables represent the quantities observed in the observer's frame (x , y , z).

For simplicity, let us first consider an axisymmetric (neutrino) source. In this case, both h_+ and $h_\times$ modes vanish for an observer on the symmetric axis, and the GW signal (h_+ component) with maximum amplitude is obtained for an observer in the equatorial plane (Müller and Janka 1997). In this case, the GW amplitude is given by

$$h_\nu = \frac{2G}{c^4 r} \int_{-\infty}^{t-D/c} dt L_\nu(t') \cdot \alpha(t'), \quad (25)$$

where $\alpha(t')$ is the time-dependent anisotropy parameter,

$$\alpha(t') = \int_{4\pi} d\Omega' \Psi(\vartheta', \varphi') f(\Omega', t'), \quad (26)$$

representing the degree of the deviation of the neutrino emission from spherical symmetry (for the concrete form of $\Psi(\vartheta', \varphi')$, see Kotake et al. 2007; Müller et al. 2012). Inserting typical values into the above equation, one can find the typical amplitudes,

$$h_\nu \sim 1.6 \times 10^2 \text{ cm} \frac{1}{D} \left(\frac{\alpha}{0.1} \right) \left(\frac{L_\nu}{10^{52} \text{ erg s}^{-1}} \right) \left(\frac{\Delta t_{\text{exp}}}{1 \text{ s}} \right), \quad (27)$$

where we take an emission time of $\Delta t_{\text{exp}} = 1 \text{ s}$ assuming constant radiation and the optimistic value of $\alpha \sim 0.1$. Thus one can expect that the GW amplitude from neutrinos can be as high as the one from quadrupole mass motions emitted near core bounce in rotational core-collapse (compare with Eq. (4)). The complete set of the GW formulae from neutrinos applicable to 3D simulations are summarized in Müller et al. (2012) and Kotake et al. (2009).

The typical GW frequency from neutrinos, on the other hand, becomes much lower ($f_{\nu, \text{GW}} \lesssim 100 \text{ Hz}$) than the one (Eq. (6)) at core bounce. Comparing with Fig. 1, one may readily see that the detection is rather hard limited by seismic noise at the low frequencies (e.g., slope $f \lesssim 100 \text{ Hz}$ in Fig. 1). As explained later, the matter GW signal at core bounce is emitted as bursts in which the wave amplitude rises from zero at core bounce, oscillates for several cycles, and then approaches zero (e.g., bottom left panel of Fig. 3). In addition, there is another class of GW, that is, *bursts with memory* (Turner and Wagoner 1979). In this case, the wave amplitude rises from zero and then settles into a nonzero final value even after the energy flows leave the source. The appearance of such an offset may be understood by looking at Δt_{exp} remaining in Eq. (25) (see an example of the waveform “with memory” in the dotted line of Fig. 8, left panel). The short time-scale modulation in the neutrino radiation field is smeared out by the memory effect, which leads to the slower variation in the $f_{\nu, \text{GW}}$.

The GWs not only generated by anisotropic neutrino emission but also from large-scale, coherent mass motions (such as jets) fall into this category (Shibata et al. 2006; Takiwaki and Kotake 2011). So the GWs from jets also have a slower temporal modulation ($\lesssim 100 \text{ Hz}$), which is hard to detect by ground-based detectors. It is noted that a future space interferometer such as the Fabry-Perot type DECIGO is designed to be sensitive in these frequency regimes (Kawamura et al. 2006). These low-frequency GW signals, if observed, could be important messengers telling us about the degree of anisotropy of neutrino emission and jet propagation in the MHD-driven explosions. It is also interesting to note that these space-based detectors could detect the GW signals from type Ia supernovae driven by a delayed detonation up to a Mpc distance scale (Seitenzahl et al. 2015).

2.3 To Be Rapidly Rotating or Not (That Is the Question!)

The gravitational waveforms of CCSNe are predominantly affected by the strength of the initial rotation rate in the precollapse iron core. Figure 2 summarizes initial angular velocity profiles from recent stellar evolutionary calculations. Very roughly speaking, “rapid rotation” corresponds to the initial angular velocity of $\Omega \gtrsim 2 - 3 \text{ rad/s}$ in the center, which is considered to be required to blow up massive stars by the MHD mechanism (e.g., Burrows et al. 2007; Mösta et al. 2014; Nishimura et al. 2015; Winteler et al. 2012). In Fig. 2, the black-dotted line, $(\Omega_0, A) = (\pi, 1000)$,

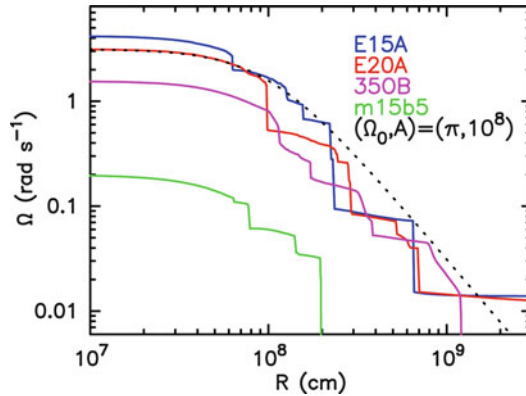


Fig. 2 Initial angular velocity (Ω) profiles as a function of stellar radii (R) from several representative progenitor models. Discontinuities in the radial profiles correspond to the composition interfaces (such as from the iron to the silicon layer for the first discontinuity, for example, at ~ 1000 km for model E20A (red line)). Models E15A and E20A are from the rotating progenitors by Heger and Langer (2000), which is thought to be the high end of Ω among all the progenitor models published thus far. Model 350B is a gamma-ray burst (GRB) progenitor, which is evolved both with rotation and magnetic fields (Woosley and Bloom 2006). Model m15b5 is a rotating progenitor with magnetic fields by Heger et al. (2005), whose initial rotation rate is estimated to be roughly compatible with that of NSs at birth (e.g., Ott et al. 2006). Two-digit numbers in each model name denote the initial progenitor mass ($15 M_{\odot}$ star is used for model E15A, e.g.)

is an analytical fit to model E20A as $\Omega(R) = \Omega_0 [1 + (\frac{R}{A})^2]^{-1}$ where A is a parameter representing the degree of differential rotation. For convenience, this angular velocity profile is referred to as “R3” (because of $\Omega_0 \sim 3$ rad/s in the center) in the following. Here it should be noted that Ω is estimated to be smaller than ~ 0.1 rad/s to explain the observed rotation periods of canonical radio pulsars (Ott et al. 2006). Therefore model m15b5 in Fig. 2 is likely one of the canonical CCSN progenitors with NS formations.

Rapid rotation (albeit still highly uncertain how rare it is; e.g., recent discussions in Fuller et al. (2015)), rapidly rotating core-collapse, and the associated GW signals are still attracting attention mainly because they are thought to connect tightly not only with hyperenergetic explosions (e.g., ► Chap. 73, “Nucleosynthesis in Hypernovae Associated with Gamma Ray Bursts”) but also with strong GW emission (e.g., nonaxisymmetric instabilities). Small initial rotation rates ($\Omega \lesssim 0.1$ rad/s) have little impact on either the dynamics or the GW emission, so that these progenitors can be regarded as nonrotating, at least in the pre-explosion phase (~ 1 s after bounce). In Sects. 3 and 4, we focus on the GW signatures from rapidly rotating and nonrotating core-collapse cores, respectively.

3 GWs from Rapidly Rotating Core-Collapse and Bounce

The left panels of Fig. 3 show the time evolution of the central density (top panel) and the gravitational waveform for a rapidly rotating model (model R3; see text above and Fig. 2). For comparison, we plot in the right panels the same quantities but for the nonrotating model (model R0). These models are computed by what are at present state-of-the-art numerical schemes (e.g., Janka et al. 2016; Kotake et al. 2012a), namely based on three-dimensional (3D) full general-relativistic (GR) simulations where the Einstein equations are evolved without approximations and with appropriate choices of temporal and spatial gauge (Baumgarte and Shapiro 1999; Shibata and Nakamura 1995). For models R0 and R3 presented below, an approximate neutrino transport scheme is also included (Kuroda et al. 2012, 2014). To maximize the GW amplitude at bounce, an observer is assumed to be located along the equatorial direction in this subsection ($\theta = \pi/2$ in Eq. (16)).

When the central density exceeds nuclear saturation density (the horizontal line in Fig. 3, $\rho_{\text{nuc}} \approx 2.8 \times 10^{14} \text{ g cm}^{-3}$), the collapsing core bounces due to the stiffening of the nuclear EOS. At this “core bounce”, the central density overshoots

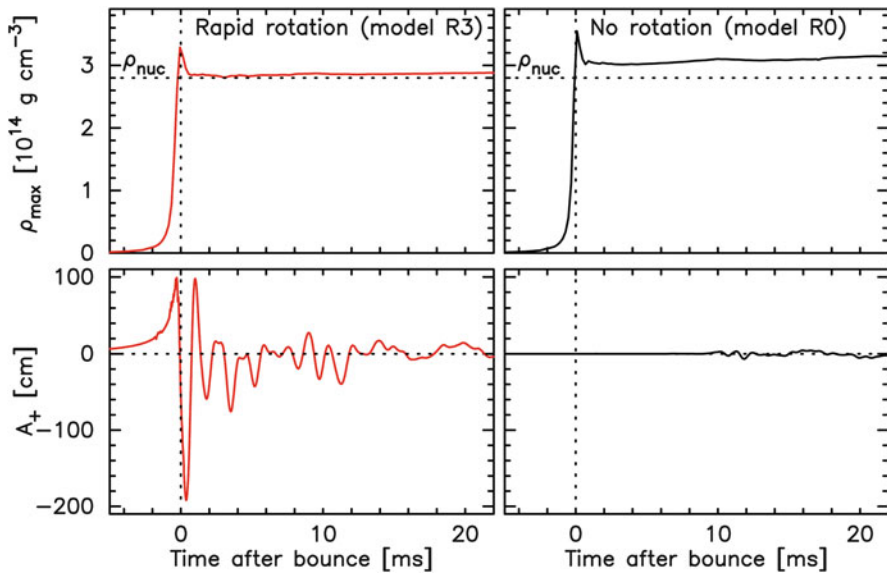


Fig. 3 Left panels show typical type I waveform (bottom panel) for a rapidly rotating core-collapse model (model R3, see text) with the time evolution of the central density (top panel). Note that A_+ is defined as $A_+ \equiv h_+ \times D$ with D representing the distance to the source (e.g., Eq. (13) for an equatorial observer (e.g., $\theta = \pi/2$)). The right panels are same as the left panels but for the nonrotating model (model R0). Note that the postbounce time (t_{pb}) is measured after the epoch of the density maximum (see text). These models are from Kuroda et al. (2014)

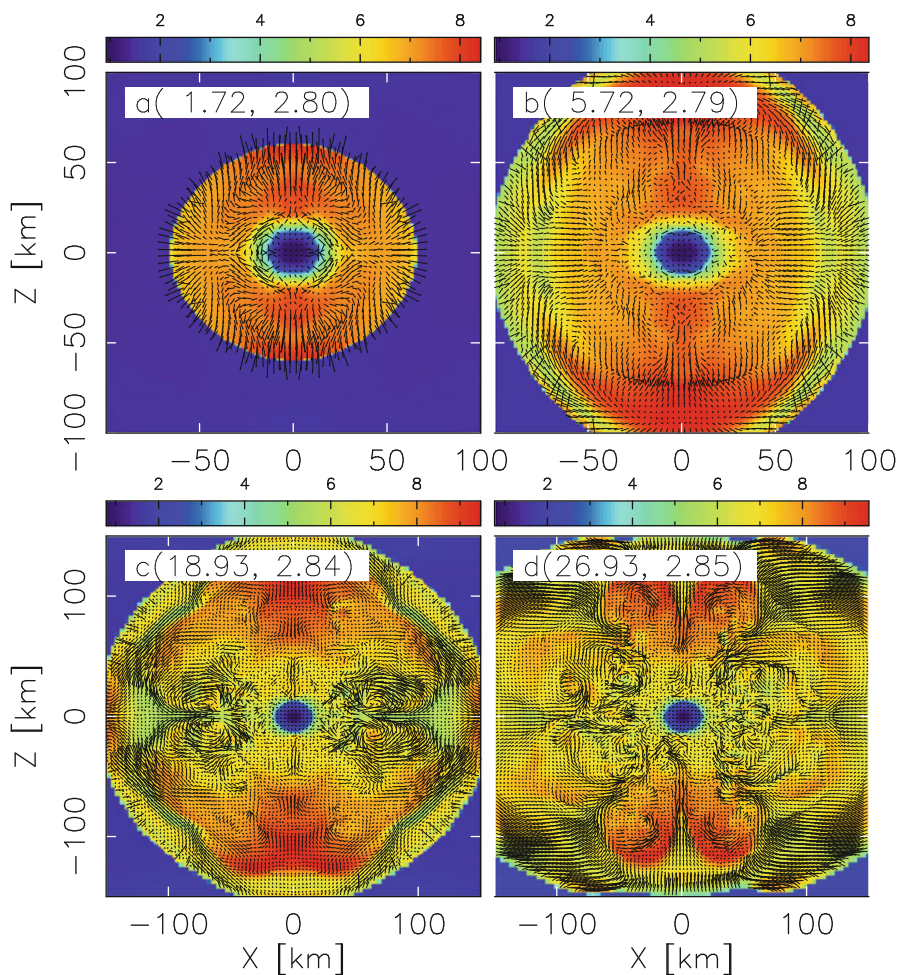


Fig. 4 Snapshots of entropy distributions showing nonspherical mass motions near core bounce. In each panel, the postbounce time (in unit of ms) and the central density (ρ_c) in units of $1 \times 10^{14} \text{ g cm}^{-3}$ are represented as (A, B) , respectively. The high-entropy postshock region is colored orange and red, which is sandwiched by the low-entropy region (colored blue) of the supersonically infalling core (exterior to the shock) and of the unshocked inner core (bluish region in the center). Normalized velocity vectors are superimposed. Note that entropy is in units of k_B^{-1} per nucleon with k_B being the Boltzmann constant. This model is from Kuroda et al. (2014)

ρ_{nuc} , then reaches its maximum (top left panel). Almost at the same time, one can see a pronounced peak in the waveform (bottom left panel). After the largest spike, weaker spikes with decreasing amplitudes appear in the waveform due to the damping oscillation of the core.

Figure 4 shows the evolution of the shock and the nonspherical mass motions in the postshock region near core bounce. First the rotating bounce occurs near the

pole (e.g., the outgoing velocity fields along $X = 0$ in panel (a) in Fig. 4). The shape of the shock is oblately deformed at this time because the material in the equatorial plane is supported by the centrifugal forces. Followed by the secondary bounce in the vicinity of the equatorial plane, the shape of the shock changes from oblate (panel (a)) to prolate (panel (b)) and then back to oblate (panel (c)). In this way, the inner core is hit by multiple downflows predominantly caused by the (damping) oblate-prolate deformation cycle in the postshock region. The typical duration of this ring-down phase is about $t_{\text{pb}} \sim 20$ ms (Dimmelmeier et al. 2007; Ott et al. 2007) as shown in the bottom left panel of Fig. 3 (see also panel (d) in Fig. 4).

The waveform (bottom left panel of Fig. 3) is categorized as “type I”, which is the most generic waveform of rotating core-collapse and bounce. The type II waveform is characterized by multiple pronounced peaks, which appears in extreme models with very rapid rotation ($\Omega \gtrsim 2\pi$ rad/s) or very strong differential rotation ($A \lesssim 500$ km) (Dimmelmeier et al. 2007; Takiwaki and Kotake 2011). When a nuclear EOS is very soft, the type III waveform is obtained, which is characterized by fast collapse, weak bounce, and small-amplitude GW emission. But recent observations of $2M_{\odot}$ NSs (Demorest et al. (2010); see Lattimer and Prakash (2016) for a review) have ruled out such soft EOSs. Therefore the generic type of the bounce GW signals is now known to take the type I waveform.

Without (rapid) rotation in the precollapse core, no distinct GW peaks are seen (e.g., bottom right panel of Fig. 3), whereas one can see small deviations ($\lesssim 10$ cm) of the GW amplitude from zero at $t_{\text{pb}} \gtrsim 10$ ms. Shortly after the bounce shock is formed, the negative entropy gradient behind the stalling shock gives rise to prompt convection. Nonspherical motions due to the prompt convection generate the GW emission, which generally lasts $t_{\text{pb}} \lesssim 50$ ms, a part of which is shown in the bottom right panel. The central density at and after bounce for the nonrotating model is higher than that for the rapidly rotating model (compare the top right with the top left panel in Fig. 3). This is because the centrifugal force acts as an additional pressure support (Mönchmeyer et al. 1991). Very rapid rotation ($\Omega \gtrsim 2\pi$ rad/s) was shown to result in very delayed core bounce at subnuclear density exclusively affected by the strong centrifugal forces (Ott 2009), possibly leading to the (rare) type II waveform mentioned above.

The top panel of Fig. 5 compares the GW spectrum for model R3 with R0, relative to the sensitivity curves of representative detectors. As one would expect, the bounce signal of model R0 is too small to be detected even for a Galactic event. The characteristic GW strain (h_c in Eq. (21)) of model R3 peaks around 1 kHz with the maximum amplitude of 2×10^{-21} , which is above the detector’s sensitivity. For a given model, the peak amplitude (h_c) and the characteristic GW frequency (f_c , Eq. (20)) are uniquely determined from the GW spectrum analysis.

The bottom panel of Fig. 5 shows a collective set of $h_c - f_c$ for multiple ($\gtrsim 100$) models, in which a wide variety of the state-of-the-art stellar evolution models is used by covering a huge parameter space spanned by initial rotation rate, degree of differential rotation, and different EOSs (Dimmelmeier et al. 2007). They demonstrated that the bounce dynamics in the rotating core and the GW signal depend primarily on the central angular velocity prior to collapse (e.g., Ω in Fig. 2).

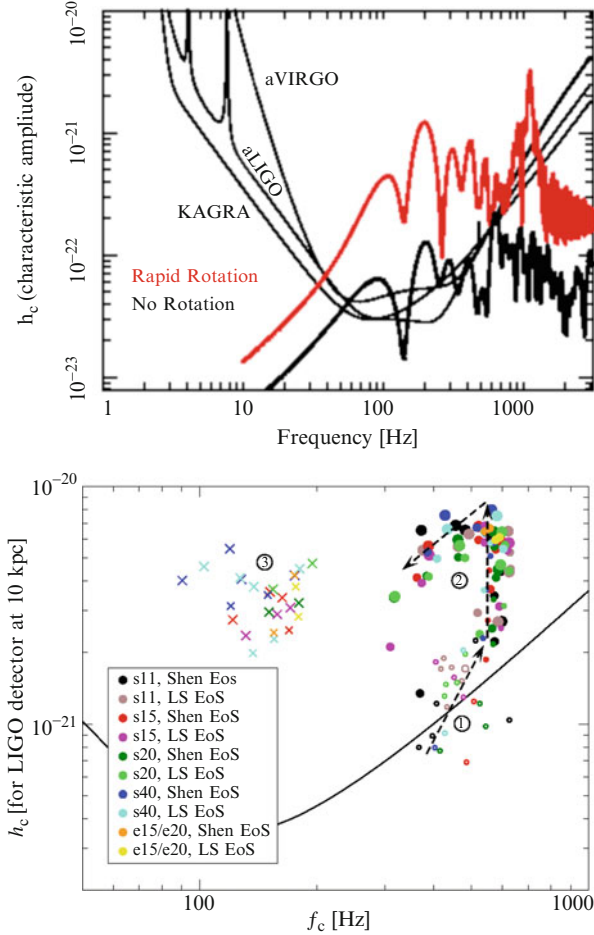


Fig. 5 *Top panel* shows spectral energy distribution of the GW signal for model R3 (red line) and R0 (thick black line) at a distance of 10 kpc with sensitivity curves of LIGO, advanced VIRGO (aVIRGO), and KAGRA, respectively (Kuroda et al. 2014). The *bottom panel* shows the location of the peak GW amplitude (h_c at 10 kpc) with the GW frequency (f_c) of $\sim$ hundred models investigated in Dimmellemeier et al. (2007). Note that each dot represents a particular model and the sensitivity of initial LIGO is found in the *bottom panel*. The three groups of signals are discussed in the text. The *bottom panel* is by courtesy of Dimmellemeier and coauthors

In fact, by looking at the bottom panel of Fig. 5 one can see that these models are basically categorized into three groups in the $h_c - f_c$ plane (Ott 2009). Slowly rotating models (group ①, $\Omega \lesssim 2 \text{ rad s}^{-1}$) have small h_c and moderately high f_c due to the weak centrifugal forces. As the impact of the centrifugal forces on the bounce dynamics becomes significant with increasing Ω , h_c becomes higher (the models move upward and slightly right in the plane). Moderately rapid models (group ②, $2 \text{ rad s}^{-1} \lesssim \Omega \lesssim 6 - 13 \text{ rad s}^{-1}$) reach high h_c and cluster in f_c between

600 and 800 Hz. The most rapidly rotating models of that group shift slightly to the left in f_c . For models with large initial angular momentum, the central density (ρ_c) after bounce can be maximally lowered by one order of magnitude compared to the nuclear density. This makes the dynamical timescale $t_{\text{dyn}} \sim (G\rho_c)^{-1/2}$ longer and the typical frequency ($f_c \propto 1/t_{\text{dyn}}$) lower. Finally very rapidly rotating models (group ③, $\Omega \gtrsim 6 - 13 \text{ rad s}^{-1}$) cluster at low f_c and lower h_c .

As one can see from the top panel of Fig. 5, the sensitivity curves in the high-frequency domain ($\gtrsim 400\text{--}1000 \text{ Hz}$) are limited by quantum noises (see a positive slope there). On the other hand, the highest detector sensitivity is roughly in the range of $\sim 50\text{--}300 \text{ Hz}$. As mentioned, a signal-to-noise ratio of 8 at least is required to claim GW detection (Eq. (22); see also Hayama et al. (2015)), (roughly) meaning that the GW amplitude should be bigger by one order of magnitude compared to the sensitivity curves. From the two panels of Fig. 5, one can expect that GWs from (moderately) rapidly rotating progenitors (groups ② and ③) in our Galaxy are a main target of current LIGO-class detectors.

3.1 Nonaxisymmetric Instabilities

As Chandrasekhar wrote in his textbook, (Chandrasekhar 1969), rapidly rotating compact stars can be subject to nonaxisymmetric rotational instabilities when the ratio of rotational to gravitational potential energy ($T/|W| = \beta$) exceeds a certain critical value. Because the growing instabilities deform the object's spheroidal into a triaxial configuration with a time-dependent quadrupole (or higher) moment, strong GW emission can be anticipated (see Fryer and New (2011) for a review).

The best-understood type of instability is the classical dynamical bar-mode instability with a threshold of $\beta_{\text{bar}} \gtrsim 0.27$. However, after a decade of extensive research on this issue (e.g., Rampp et al. 1998), recent work (e.g., right panel of Fig. 5) presented strong evidence that the postbounce core, even in extreme models, does not reach values of β close to β_{bar} during collapse, bounce, and early postbounce phase. Regarding the various types of instabilities that would accompany strong GW emission in the PNS cooling phase (lasting on a timescale of $\sim$ minutes; see e.g., ► Chap. 60, “Neutrino Signatures from Young Neutron Stars”) and afterwards, the reader is referred to Kokkotas and Schmidt (1999).

In contrast to the high $T/|W|$ instabilities, recent work suggests that a differentially rotating PNS can become dynamically unstable at much lower $T/|W|$ as low as $\lesssim 0.1$ (Kuroda et al. 2014; Ott et al. 2005; Scheidegger et al. 2010). The origin of the “low- $T/|W|$ instability” is not completely understood yet. However, the standard picture (e.g., Watts et al. 2005) was given, in that the instabilities are associated with the existence of corotation points (where the pattern speed of the unstable modes matches the local angular velocity) inside the core and are thus likely to be a subclass of shear instabilities.

Figure 6 shows several hydrodynamics and GW features for a representative 3D MHD model that experiences the low- T/W instability in the postbounce phase (Scheidegger et al. 2010). The top left panel shows the vorticity distribution in the

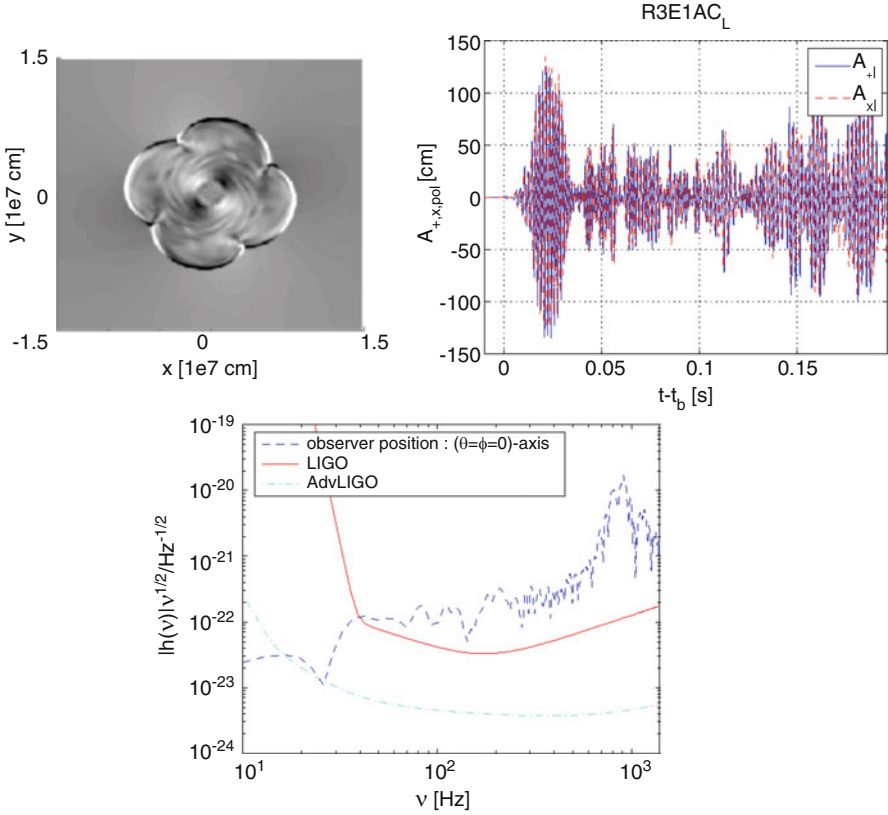


Fig. 6 *Top left panel* depicts the vorticity distribution in the equatorial plane of a representative 3D MHD model (R4E1FC_L) in Scheidegger et al. (2010) (at ~ 10 ms after bounce). A two-armed $m = 2$ pattern behind the stalled shock is clearly seen (the side length of the plot is $(300 \text{ km})^2$). The *top right panel* displays the quadrupole GW amplitude of the A_+ and $A_\times$ (see text for the definition) along the pole (of a model (R3E1FC_L) showing a similar hydrodynamics evolution to the *left panel*). The *bottom panel* shows the spectral energy distributions of the GW signal for their representative 3D model experiencing the low- $T/|W|$ instability (model R4STCA) at a distance of 10 kpc. The observer is assumed to be along the polar axis. These plots are by courtesy of Scheidegger and coauthors

equatorial plane. A two-armed ($m = 2$ with m denoting the azimuthal quantum number) pattern is clearly seen (at ~ 10 ms after bounce for the plot) and then the spiral flows develop for more than several hundred milliseconds later. The right panel displays the gravitational waveform of the A_+ and $A_\times$. Note that $A_{+,x}$ is defined as $A_{+,x} \equiv h_{+,x} \times D$ with D representing the distance to the source (e.g., Scheidegger et al. (2010) for more detail). To maximize the amplitude, the observer is assumed to be located along the spin axis of the star. An important message from the top panel is that the gravitational waveform from the nonaxisymmetric dynamics generally shows narrowband and highly quasi-periodic signals that persist until the end of simulations. For the periodic signals, the effectively measured GW amplitude

scales with the number of GW cycles N as $h_{\text{eff}} \propto h\sqrt{N}$. This is the reason why the peak amplitude in the GW spectrum (the bottom panel in Fig. 6) exceeds 10^{-20} for the galactic source, making the chance of detection quite higher. In fact, the wave amplitude from the short-duration bursts near bounce is generally smaller than 10^{-20} for the galactic source (e.g., the amplitude near 1 kHz in the bottom panel in Fig. 6).

For detecting (potentially) the most powerful GW signals, it is therefore of crucial importance to understand the properties of the nonaxisymmetric instabilities. However, the numerical difficulty in following a long-term postbounce evolution in 3D is the main hindrance at present. Although angular momentum is continuously brought into the iron core from the stellar envelope that rotates with higher angular momentum, the shear energy in the vicinity of the corotation points will continue to be redistributed or even dissipated by some mechanisms including the magnetorotational instability (MRI, e.g., Akiyama et al. 2003; Masada et al. 2012). In the 3D model presented in Fig. 6, neutrino heating was not taken into account. Neutrino-driven convection would affect the growth of the nonaxisymmetric instabilities. GR, which has been mostly treated in 3D models (Hanke et al. 2013; Lentz et al. 2015) by a post-Newtonian manner, should affect the growth rate. For a more quantitative discussion, full 3D GR-MHD simulations with an appropriate neutrino transport are needed, towards which supernova modelers are making their best efforts (Janka et al. 2016 for a review).

4 GWs from Nonrotating Progenitors

With decreasing the precollapse angular velocity, the pronounced “type I” waveform of Sect. 3 also becomes weak (compare the bottom left to the bottom right panel in Fig. 3). For slowly rotating progenitors ($\Omega \lesssim 0.1$ rad/s, Fig. 2), the bounce signal is well below the sensitivity of the LIGO-class detectors even for a Galactic event. The first sizable GW signal after bounce comes from prompt convection as we briefly mentioned in panel (d) of Fig. 4. The growth of “convection” is naturally governed by the nonlinear hydrodynamics. As we describe in this section, the gravitational waveforms in the postbounce phase are all affected by turbulent mass motions that are originated from various types of fluid instabilities. Therefore the waveforms in the postbounce phase are of stochastic nature (Kotake et al. 2009) and impossible to predict a priori.

The most established method to detect the GW signal is a matched filtering (see Sathyaprakash and Schutz 2009 for review) as is done when one looks for compact binary coalescence signals (e.g., Rasio and Shapiro 1999), leading to the first detection of the GW150914 event (Abbott et al. 2016b). However, such a template-based search is not suitable for the GW signals from CCSNe due to the stochastic nature. This makes the detection even harder, however, there is some hope. A characteristic GW frequency (Müller et al. 2013, 2004; Murphy et al. 2009) is expected to provide an important probe to multi-D hydrodynamics episodes in the postbounce core. From the next section, we move on to focus on the postbounce GW signatures.

4.1 Characteristic GW Frequency in the Postbounce Phase

The left panel of Fig. 7 shows a sample of gravitational waveforms obtained in 2D models by Murphy et al. (2009). Shortly after the bounce shock is formed, a negative entropy gradient behind the stalling shock predominantly gives rise to prompt convection. GWs indicated by “prompt convection” come from this. Later, the PNS convection driven by the negative lepton gradient near its surface and the neutrino-driven convection in the postshock heating region develop. This corresponds to the GW emission by “postshock convection” after the prompt convection till ~ 550 ms after bounce. Subsequently, the GW amplitudes become much larger as the SASI enters a nonlinear phase with violent sloshing of the postshock material. Large spikes appearing in the “Nonlinear SASI” phase come from the downflowing “SASI plumes” striking the PNS surface. Afterward when the sloshing shock develops into an explosion (indicated by “Explosion”), the sign of the GW amplitudes changes, reflecting the geometry of the expanding shock, that is either prolate (increase), oblate (decrease), or spherical (in-between; see Murphy et al. 2009 for more detail).

Interestingly the f_p curve (black solid line) closely traces the three excesses in the GW spectrogram. The strong correlation implies that the characteristic GW frequency is considered as a result of the deceleration of convective plumes hitting the surface (Murphy et al. 2009) and by the g -mode oscillation excited by the downflows to the PNS (Marek et al. 2009; Müller et al. 2013). Following Müller et al. (2013), the characteristic g -mode frequency in the vicinity of the PNS can be estimated as

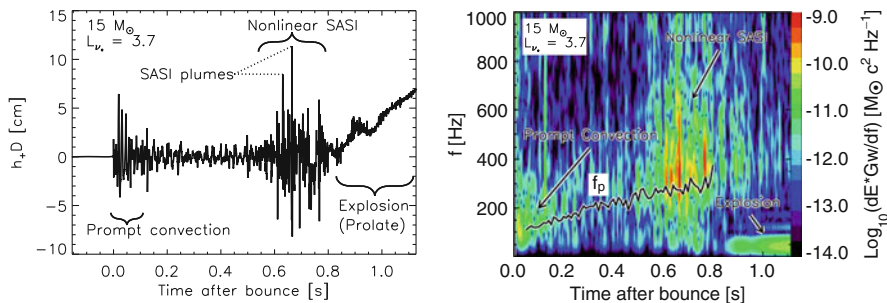


Fig. 7 The *top left panel* shows a gravitational waveform ($h_+ \times D$ with D representing the distance to the source, for example, Eq.(16)) obtained in a 2D hydrodynamics model for a $15 M_\odot$ star (Murphy et al. 2009). Three typical GW emission sites are indicated, which are prompt convection, nonlinear SASI with SASI plumes(then till ~ 800 ms after bounce), and explosion (then afterwards). For the calculation presented here, an input neutrino luminosity of $L_{\nu_e} = 3.7 \times 10^{52}$ erg/s was assumed to trigger neutrino-driven explosions. The *right panel* shows a GW spectrogram dE_{GW}/df (Eq. (18)) as a function of the postbounce time. The characteristic GW frequency “ f_p ” is superimposed. Comparing with the *left panel*, there are three features showing an excess in the spectrogram (green, orange, red regions), which correspond to prompt convection, nonlinear SASI/convective motions, followed by the explosion phase, respectively. These plots are by courtesy of Murphy and coauthors

$$f_p = \frac{1}{2\pi} \frac{GM_{\text{pns}}}{R_{\text{pns}}^2} \sqrt{\frac{(\Gamma - 1)m_n}{\Gamma k_B T_{\text{pns}}}} \left(1 - \frac{GM_{\text{pns}}}{R_{\text{pns}} c^2}\right), \quad (28)$$

where M_{pns} , R_{pns} , and T_{pns} are the PNS mass, radius, and (typical) temperature, respectively; Γ is the adiabatic index in the PNS surface region; m_n is the neutron mass; and $(1 - \frac{GM_{\text{pns}}}{R_{\text{pns}} c^2})$ is the GR correction. Mass accretion to the PNS predominantly makes M_{pns} bigger, R_{pns} smaller, and T_{pns} higher, whereas other important ingredients (neutrino cooling, GR) should make a nontrivial contribution to the above trend. The former two are primarily responsible for the increasing trend of f_p in the spectrogram. (See also Marek et al. 2009; Müller et al. 2012; Yakunin et al. 2015.)

The top left panel of Fig. 8 shows a gravitational waveform obtained in 2D GR core-collapse simulations of a $15M_{\odot}$ star where currently one of the best available

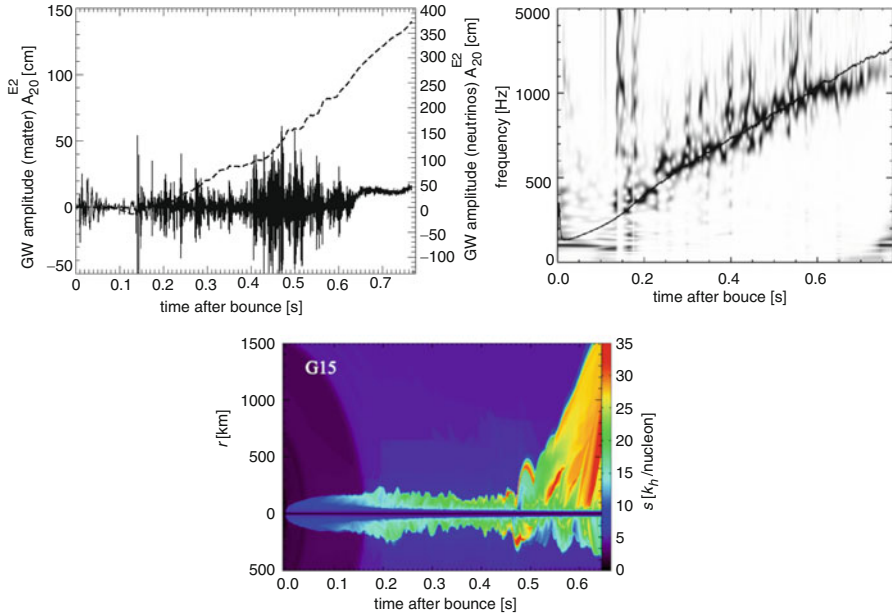


Fig. 8 Top left panel shows gravitational waveform obtained by state-of-the-art 2D simulations for a $15 M_{\odot}$ star (model G15 in Müller et al. (2013)). Solid and dashed lines represent matter and neutrino GW signals (e.g., Eqs. (16) and (25)), respectively. Note that the scale for the matter GW (left vertical axis) is different from that for the neutrino GW signal (right vertical axis). The top right panel is the GW spectrogram overlaid by the f_p formula (solid black line, Eq. (28)). On top of the monotonically increasing f_p curve, the high-frequency components are superimposed (vertical stripes extending up to $\gtrsim 1$ kHz) that come from the strong downdrafts stochastically hitting the PNS surface region. The bottom panel shows the entropy along the north and south polar axis as a postbounce bounce time. The shock revival was obtained for model G15 around ~ 500 ms postbounce, followed by the runaway expansion trending towards explosion more strongly towards the north pole. These plots are by courtesy of Müller and coauthors

neutrino transport schemes is employed (Müller et al. 2013). As mentioned in Sect. 2.2.1 (Eq. (25)), the GW amplitude from anisotropic neutrino emission (dashed line, denoted “neutrino GW” in the caption) becomes higher than that of matter quadrupole GW (solid line). The prompt convection GW signal is seen $t_{\text{pb}} \lesssim 50$ ms, which is followed by a quiescent phase until $t_{\text{pb}} \sim 150$ ms. As neutrino-driven convection and the SASI vigorously develop with time the matter GW amplitude and the characteristic GW frequency f_p become higher with time (solid line in the top right panel). These basic GW features have been observed not only in other self-consistent 2D models (Marek et al. 2009; Yakunin et al. 2015), but also in 3D simulations with idealized neutrino transport schemes (Müller et al. 2012).

It should be noted that one needs 3D-GR models with an appropriate neutrino transport scheme (e.g., Kuroda et al. 2016) to unambiguously determine the postbounce GW features. The final goal is to develop a full 3D GR-MHD code with Boltzmann neutrino transport (e.g., Janka et al. (2016) and Kotake et al. (2012a)) including state-of-the-art EOS (e.g., Hempel and Schaffner-Bielich 2010) and matter-neutrino interactions (e.g., Langanke and Martínez-Pinedo 2003; e.g., ► Chap. 70, “Neutrinos and Their Impact on Core-Collapse Supernova Nucleosynthesis”). Unfortunately none of the currently published CCSN models satisfies the “ultimate” requirement. In this sense, all the multi-D models employ some approximations (with different levels of sophistication) for multiple physical ingredients. In the same way, theoretical predictions of the GWs that one can obtain by analyzing the currently available numerical results, cannot unambiguously give us the final answer yet. Again we note that this chapter shows only a snapshot of the moving theoretical terrain towards the final goal.

5 Conclusions

The aim of writing this chapter was to summarize the baseline physics for understanding the GW signatures in stellar core-collapse and CCSN evolution. In Sect. 1, we gave a quick introduction to overview how we could detect scientifically significant GW signals from CCSNe in the future. In Sect. 2, we presented a concise summary of the formulae to extract gravitational waveforms from CCSN models (Sects. 2.1 and 2.2). By using them, we did a back-of-the-envelope estimate for understanding the fundamental features of the GWs and for discussing (first-order) detectability by the LIGO-class detectors.

Section 2.3 emphasized that the precollapse rotation rate in the iron core predominantly determines the GW signatures of CCSNe. In Sect. 3, we focused on the best-studied GW signal that is emitted from rapidly rotating core-collapse and bounce. Using results from the most recent 3D GR models, we explained in detail the properties of the most generic “type I” waveform. After that, we touched on the GW signatures from nonaxisymmetric instabilities not only because they can be potentially the strongest GW emission process, but also because future detection of circular polarization in the GW signal would provide strong evidence of rapid rotation in the precollapse cores (Hayama et al. 2016).

In Sect. 4, we switched gears to focus on GWs from nonrotating progenitors. In contrast to the GW at bounce, multiple GW emission processes have been proposed and studied extensively in the last decade. Among them, we mainly focused on the GW signatures from prompt convection, neutrino-driven convection, and the SASI in the postshock region. These postbounce GWs are generated by turbulent matter motions governed by nonlinear hydrodynamics; thus they are of stochastic nature. For these stochastic GWs, a (template-based) matched filtering method cannot be applied, which makes the detection even harder. Be that as it may, we discussed that the characteristic GW frequency (e.g., Eq. (28)) can be an important probe to decipher the hydrodynamics episodes and the PNS evolution in the postbounce phase.

To have the final words on these postbounce GW signatures, we need, at least, 3D GR-MHD models where appropriate neutrino transport schemes are taken into account (e.g., Kotake et al. (2012a) and Janka et al. (2016) for a status report). As repeatedly mentioned elsewhere in this chapter, our understanding of the GW emission processes in CCSNe has been progressing in accordance with the sophistication of numerical simulations into which SN modelers have been putting a huge effort for a long time. What we have attempted to provide here is only a snapshot of the moving documentary film that records our endeavors for living out our dream of “GW astronomy of CCSNe.”

Now that the first successful GW detection revealed the importance of a BH binary as a GW source (Abbott et al. 2016b), the documentary film is becoming even longer. This is because we have to clarify the dynamics of CCSNe with a BH formation, which is nevertheless still very uncertain at this stage. After the first GW detection, multimessenger astronomy is now becoming a reality where a synergic analysis using GWs, neutrinos, and electromagnetic messengers should be indispensable for deciphering the central engines of massive stars (e.g., Leonor et al. 2010; Nakamura et al. 2016).

Needless to say, it is of crucial importance to determine nucleosynthesis accurately (see also ► Chap. 71, “Making the Heaviest Elements in a Rare Class of Supernovae”) in the SN ejecta for modeling supernova light curves (e.g., Thielemann et al. (2011), e.g., ► Chap. 68, “The Multidimensional Character of Nucleosynthesis in Core-Collapse Supernovae”). Detailed spectroscopic observations will reveal the explosion geometry (Grefenstette et al. 2014), which should constrain the asphericity of the central engines. Here we stop penning this chapter hoping that in the near future great progress will be brought by you who are reading this, in a number of unsettled and exciting riddles raised in this chapter,

6 Cross-References

- Detecting Gravitational Waves from Supernovae with Advanced LIGO
- Making the Heaviest Elements in a Rare Class of Supernovae
- Neutrino Emission from Supernovae
- Neutrinos and Their Impact on Core-Collapse Supernova Nucleosynthesis
- Neutrino Signatures from Young Neutron Stars

- ▶ [Nucleosynthesis in Hypernovae Associated with Gamma Ray Bursts](#)
- ▶ [The Multidimensional Character of Nucleosynthesis in Core-Collapse Supernovae](#)

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